Lasso Regression

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Where this fits: Pipeline map step 8 (Model). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
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- **Builds on:** Feature selection; Lagrange multipliers, KKT and duality; MAP estimation; Regularisation.
- **Leads to:** Elastic Net.
- **Compare with:** Ridge regression; L1 and L2 regularisation in neural networks.

1. Overview

Key point: Lasso is Ridge with the squares in the penalty replaced by absolute values. That one change lets it push coefficients to exactly 0, so it also removes useless **features** (G-772; input variables, the columns of the data table): automatic feature selection.

Lasso regression (G-1047) is the second regularised version of linear regression, also called **L1 regularisation** (G-1026). Like Ridge (the idea of penalising large coefficients), it adds a penalty to the usual squared error, but the penalty uses the absolute size of each coefficient. A **coefficient** is the number that multiplies a feature in the model, such as the slope m in $y = mx + b$ (see the equation). In words: the loss is the usual sum of squared errors, plus λ times the sum of the sizes of the coefficients. For two coefficients 3 and -2 the size-sum is:

$$|3| + |-2| = 3 + 2 = 5$$

Ridge's square-sum would be:

$$9 + 4 = 13$$

Written as a formula:

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{m} |\beta_j|$$

	Ridge	Lasso
Penalty	$\lambda \sum \beta_j^2$	$\lambda \sum \lvert\beta_j\rvert$
Also called	L2 regularisation	L1 regularisation
Coefficients for large λ	close to 0, never exactly 0	exactly 0

The symbols: y_i is the actual target of observation i, $\hat{y}_i$ the prediction, n the number of observations, m the number of features, β_j the coefficient of feature j (for the diabetes data, $m = 10$), and λ the penalty strength. LASSO stands for "least absolute shrinkage and

selection operator" (Tibshirani 1996). As with Ridge, $\lambda \geq 0$: with $\lambda = 0$ Lasso is plain linear regression, a small λ may overfit, and a large one underfits. The intercept is not penalised, because it only sets the average level of y, and shrinking it would not make the line flatter (the idea of Ridge).

This Note shows what Lasso does with one feature, with a flexible polynomial, and with the 10-feature diabetes data (442 **observations** (G-1374), that is, records or rows; the **target** (G-1949) we predict is disease progression; the features are age, sex, bmi (body mass index), bp (blood pressure) and six blood measurements s1 to s6), and why its coefficients can reach exactly 0.

2. One feature: the slope reaches exactly 0

> **Key point:** As alpha grows, the Lasso slope falls in a straight line and stops at exactly 0.

Figure 1 fits Lasso to the 100-observation example of the Ridge Notes. scikit-learn calls the penalty strength `alpha`.

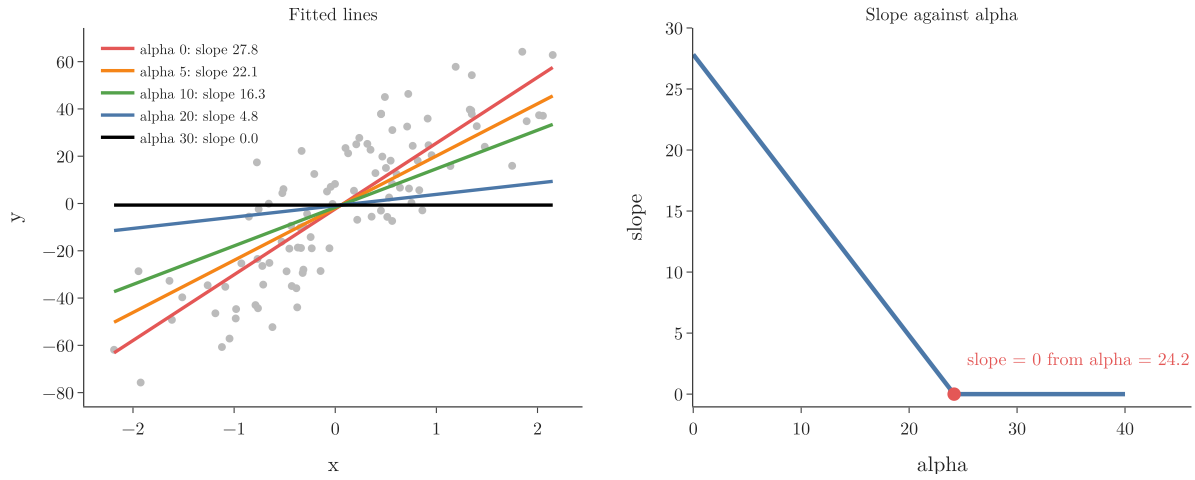


Figure 1: Lasso with one feature: fitted lines and the slope against alpha

alpha	Slope	Intercept
0 (linear regression)	27.83	−2.29
5	22.07	−1.96
10	16.31	−1.62
20	4.80	−0.95
25 and above	0	−0.67

From alpha 24.17 on, the slope is exactly 0. The line is then flat at the average of y (−0.67):

the feature no longer plays any part. With Ridge, the slope only approached 0 (see why the slope shrinks).

> **Extra:** scikit-learn's `Lasso` minimises $\frac{1}{2n}\sum(y_i - \hat{y}_i)^2 + \alpha \sum |\beta_j|$ (scikit-learn docs, `Lasso`): it averages the squared error and halves it. So its `alpha` is on a different scale from the λ of the formula above, and from `Ridge`'s `alpha`. The values of alpha for Ridge and Lasso cannot be compared directly.

3. A flexible model: Lasso picks the right terms

> **Key point:** On a degree-16 polynomial, a moderate alpha set 14 of the 16 coefficients to 0 and kept exactly x and x^2, the true shape of the data.

Figure 2 fits a degree-16 polynomial to 100 points made from this curve plus noise:

$$y = 0.7x^2 - 2x + 3$$

The 16 power columns (x to x^{16}) are standardised first (rescaled to mean 0 and spread 1; see the standardization formula).

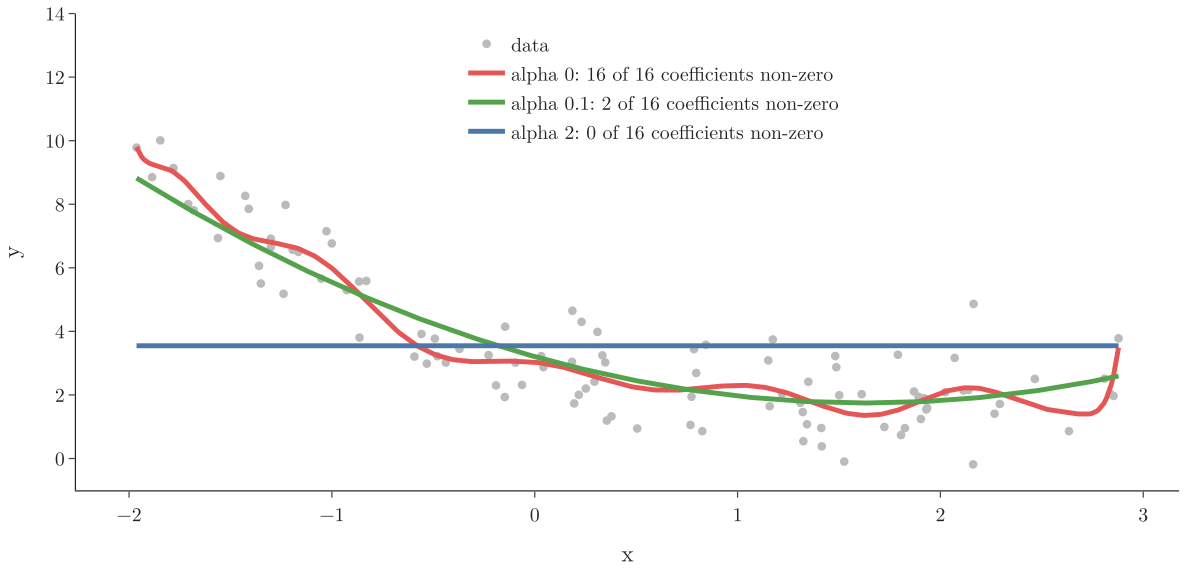


Figure 2: Degree-16 polynomial with Lasso for three values of alpha

- **alpha 0** (red): all 16 coefficients are used, and the curve wiggles to follow the noise, swinging up at the right edge: overfitting.
- **alpha 0.1** (green): only 2 coefficients are non-zero, those of x and x^2. The curve is a smooth parabola, the true shape.
- **alpha 2** (blue): every coefficient is 0, and the model predicts the average everywhere: underfitting.

alpha	Non-zero coefficients
0.01	x, x^2, x^3, x^{16}
0.1	x, x^2
1	x

Lasso found which of the 16 power features matter without being told.

4. Feature selection

> **Key point:** Coefficients that reach 0 remove their features from the model. So Lasso selects features while it trains.

A coefficient of exactly 0 means that feature has no effect on the predictions, so its column can be dropped. A model in which many coefficients are exactly 0 is a **sparse model** (G-1844). Zero coefficients make Lasso a tool for **feature selection** (G-768; see the feature engineering), useful when there are many features and some of them do not matter. Think of packing a suitcase with a strict weight limit: the items you barely need are left out entirely.

Figure 3 trains Lasso on the diabetes data (test size 0.2, random state 2). The last column of the table below is the test R^2, the share of the **target** (G-1949)'s variation the model explains on the test data: 1 is perfect, 0 is no better than predicting the average (see **R^2 score** (G-1717)).

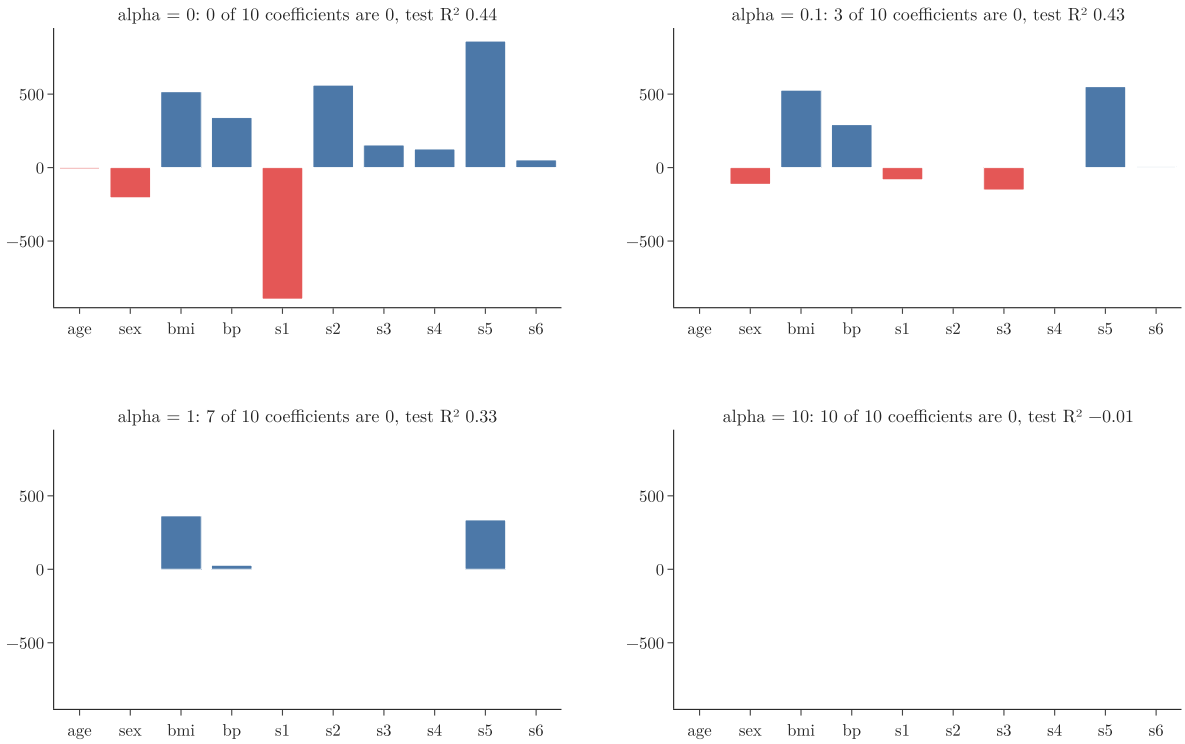


Figure 3: Diabetes coefficients for alpha 0, 0.1, 1 and 10, on the same scale

alpha	Coefficients at 0	Features kept	Test R^2
0	0	all 10	0.44
0.1	3	7 (age, s2, s4 dropped)	0.43
1	7	bmi, bp, s5	0.33
10	10	none	−0.01

At alpha 0.1, Lasso drops three features and loses almost no accuracy (0.43 against 0.44). At alpha 1, Lasso keeps only bmi, bp and s5, the three most useful features here. At alpha 10, Lasso removes everything: underfitting.

Ridge cannot do this. Ridge shrinks an unhelpful coefficient to, say, 0.2, but the feature stays in the model.

A test with useless features. Suppose we predict a mouse's size from its weight, its diet, its astrological sign and the airspeed of a swallow. The last two are useless. Ridge would shrink their coefficients but keep them in the equation; Lasso can remove them (StatQuest, "Regularization Part 2"). Figure 4 runs this test on the diabetes data. We add 10 columns of pure random noise, on the same scale as the 10 real features, and fit both models.

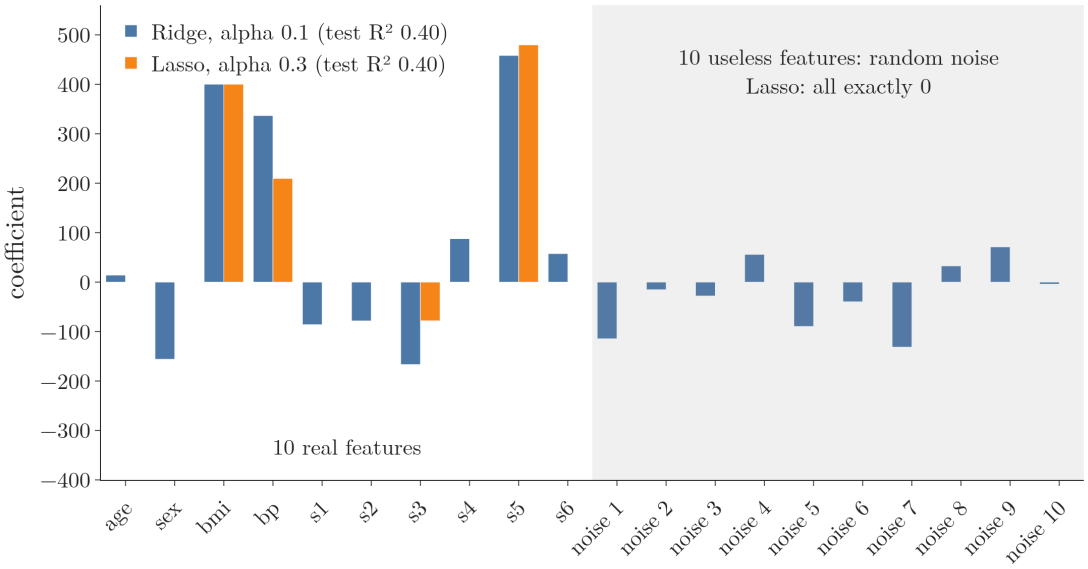


Figure 4: Coefficients on the diabetes split with 10 extra columns of random noise (shaded). Ridge with alpha 0.1 (blue) gives every noise column a non-zero coefficient. Lasso with alpha 0.3 (orange) sets all 10 noise coefficients to exactly 0 and keeps bmi, bp, s3 and s5. Both score test R^2 0.40.

- **Ridge** (alpha 0.1) gives all 10 noise columns a coefficient, some as large as −131.
- **Lasso** (alpha 0.3) sets all 10 to exactly 0 and keeps four real features: bmi, bp, s3 and s5.
- Both reach test R^2 0.40, against 0.38 for plain linear regression on the 20 columns.

The result does not depend on one lucky draw of noise: over 100 different draws, Lasso zeroes 9.65 of the 10 noise columns on average, and Ridge none (script `useless.py`).

> **Extra:** Why call bmi, bp and s5 the most useful three? Among all 120 possible sets of three features, these three give plain linear regression the best cross-validated R^2 (0.485).

> **Extra:** Feature selection is why Lasso is often preferred when the data has many columns, some of which are probably irrelevant: Lasso tends to do best when only a few features really matter (ISL §6.2.2). Fewer features also means a simpler model that is easier to explain (ISL §6.2.2).

5. How the coefficients change

> **Key point:** The coefficients hit 0 one at a time. The size of a coefficient does not decide when it goes: the large s2 goes second, while bmi and s5 survive longest.

Figure 5 follows each coefficient as alpha grows on a log scale (each tick mark on the alpha axis is ten times the one before).

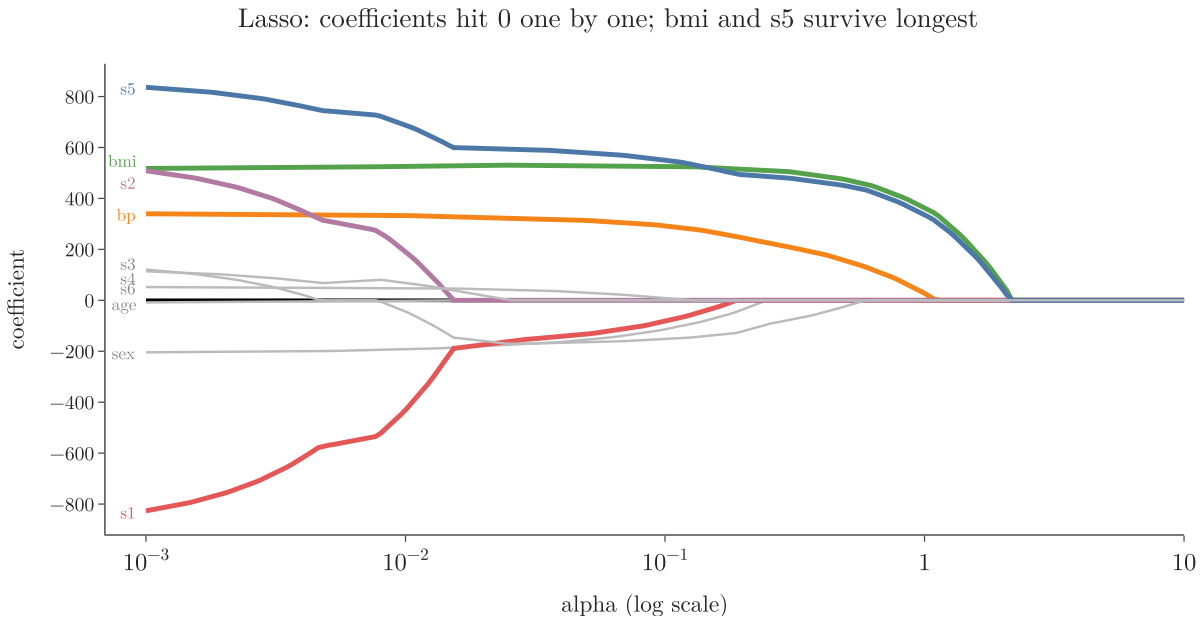


Figure 5: Lasso coefficient paths on the diabetes data

Feature	Coefficient at alpha 0	Exactly 0 from alpha about
age	−9	0.012
s2	561	0.015
s4	127	0.026
s1	−896	0.20
bp	341	1.1
bmi, s5	517, 861	2.2

- As with Ridge, the very large coefficients (s1, s2, s5) are cut back hard at first. The cut barely changes the test R^2 (0.440 at alpha 0, 0.441 at alpha 0.01, 0.433 at alpha 0.1).
- Then the coefficients drop to 0 one after another. bmi and s5 are the last to go.
- s1 and s2 had huge coefficients (−896 and +561) because they are strongly correlated with each other ($r = 0.90$): such a pair can get large opposite coefficients that partly cancel out (ESL §3.4.1). Lasso drops them early anyway: size alone does not protect a coefficient.

6. Bias and variance

> **Key point:** As with Ridge, a larger alpha raises bias and lowers variance. The best alpha is in between.

We use the same method as the Ridge bias-variance method: made-up data from the known curve plus noise (standard deviation 2), 20 fixed training **observations** (G-1374), a degree-16 polynomial, and 200 fresh noise draws. Bias^2 (how far the average fit is from the true curve)

and variance (how much the fits differ from draw to draw; see bias and variance) are measured against the true curve at 19 test points between the training ones.

alpha	Bias^2	Variance	Expected test error
0.001	0.009	1.44	5.45
0.01	0.009	0.95	4.96
0.1	0.04	0.62	4.66
0.3	0.34	0.54	4.88
1	2.52	0.38	6.90
3	5.20	0.19	9.40

Figure 6 shows the experiment. At a tiny alpha the 200 fitted curves fan out, most of all at the edges: high variance. At alpha 3 every fit is a flat line at the average, the same for every draw: almost no variance, but far from the true curve. At alpha 0.1 the fits hug the true parabola.

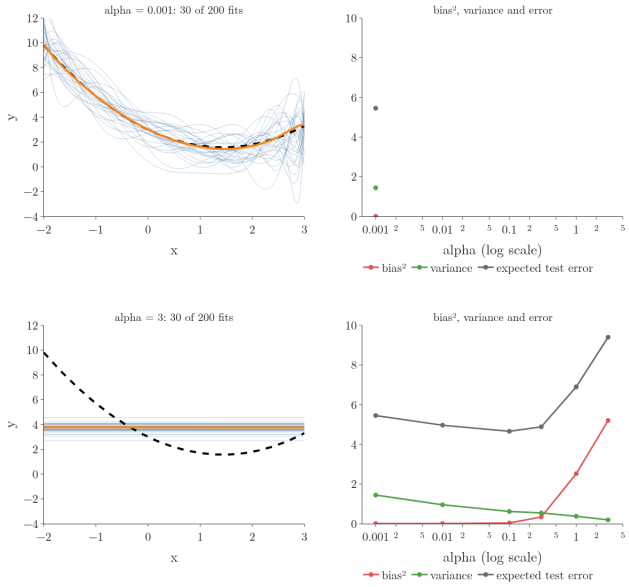


Figure 6: Bias and variance of Lasso on a degree-16 polynomial. Left: 30 of the 200 fitted curves (blue), their average (orange) and the true curve (dashed) for each alpha. Right: bias^2, variance and expected test error, built up alpha by alpha; the error is lowest at alpha 0.1.

Small alpha: high variance (**overfitting**, G-1429). Large alpha: high bias (**underfitting**, G-2035). Here alpha 0.1, the same value that kept only x and x^2 in Section 3, has the lowest total error.

> **Extra:** Expected test error = bias^2 + variance + noise (ISL §2.2.2, eq. 2.7); the noise part is $2^2 = 4$, the floor no model can beat.

7. Why coefficients reach exactly 0

> **Key point:** The penalty $\lambda|\mathrm{m}|$ makes a sharp corner in the loss at $\mathrm{m} = 0$. Once λ is large enough, that corner is the lowest point, and it stays there however large λ gets.

Take the one-feature example again and hold the intercept at -2.29, so the loss depends only on the slope:

$$L(m) = \sum_{i=1}^{n} (y_i - mx_i + 2.29)^2 + \lambda|m|$$

Figure 7 draws this curve while λ grows from 0 to 8,000.

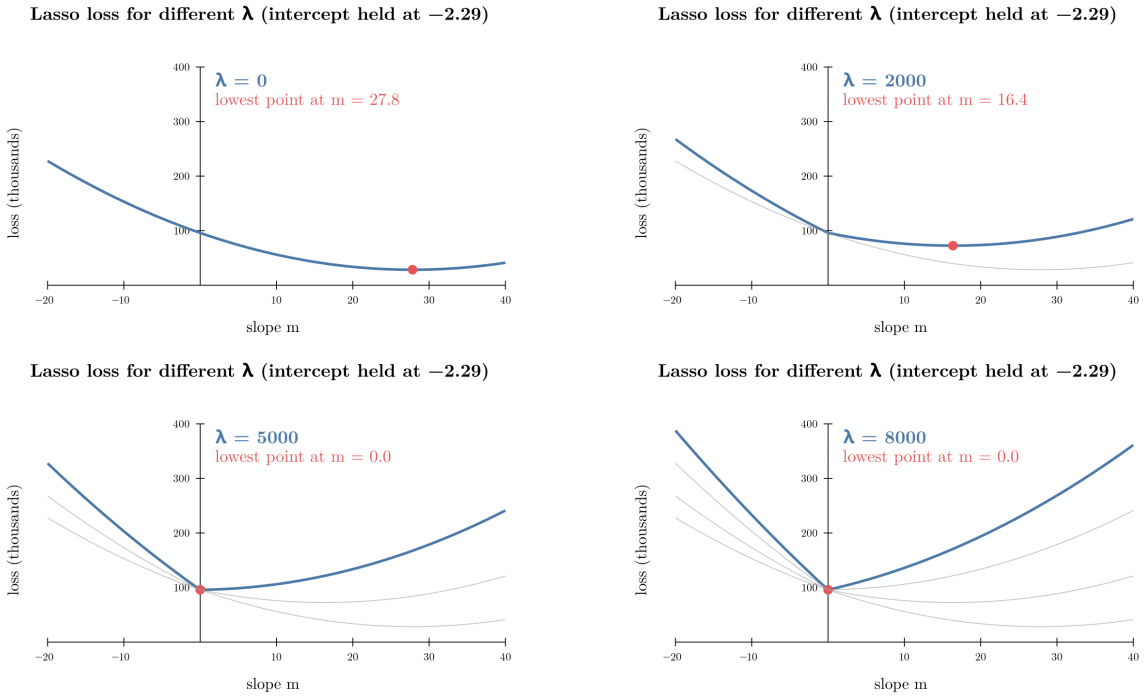


Figure 7: The Lasso loss against the slope for $\lambda = 0$, 2000, 5000 and 8000

λ	Slope at the lowest point
0	27.8
2,000	16.4
5,000	0
8,000	0

- The penalty $\lambda|m|$ is a V shape: it is 0 at $m = 0$ and grows at the same rate on both sides. Added to the smooth bowl, it gives the curve a sharp corner at $m = 0$.
- As λ grows, the lowest point moves left towards 0, as with Ridge.

- Once λ passes about 4,850 here, the corner itself is the lowest point. From then on the answer is exactly $m = 0$, and a larger λ only makes the corner steeper.

The Ridge penalty λm^2 is smooth and flat at $m = 0$, with no corner, so its lowest point only approaches 0 (see Ridge coefficients shrink but never reach 0). The next Note derives this precisely.

Figure 8 puts the two loss curves next to each other, with the same λ in both panels. Watch the red dots. On the left, the Ridge curve stays a smooth bowl; its lowest point glides towards 0 and is still at 0.30 when $\lambda = 8{,}000$. On the right, the Lasso curve grows a corner at $m = 0$, and from $\lambda = 5{,}000$ the lowest point sits exactly on that corner.

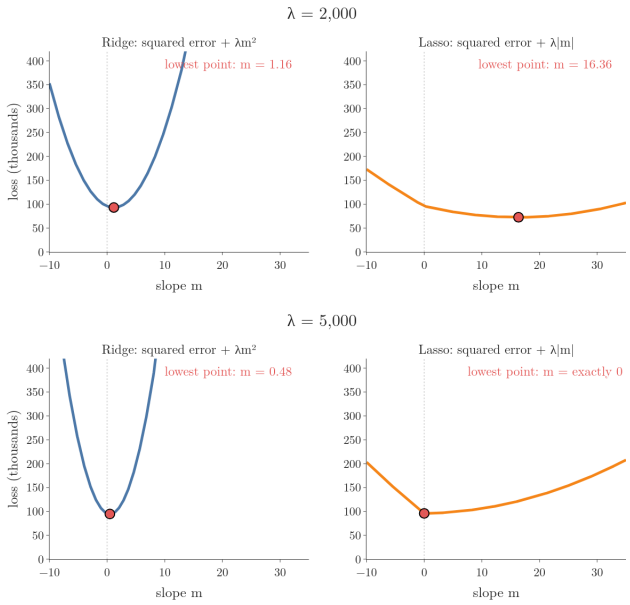


Figure 8: The Ridge loss (left) and the Lasso loss (right) against the slope, for the same λ from 0 to 8,000. The red dot marks the lowest point: Ridge's glides towards 0 without reaching it; Lasso's reaches exactly 0 at $\lambda = 5{,}000$ and stays. Idea after StatQuest, "Ridge vs Lasso Regression, Visualized!!!"

> **Extra:** Why do the two penalties behave so differently near 0? Look at how hard each penalty pushes, which is its slope. The slope of λm^2 is $2\lambda m$: at $m = 0.1$ it is only 0.2λ, at $m = 0.01$ only 0.02λ, and it fades to 0 as m approaches 0, so Ridge stops pushing. The slope of $\lambda|m|$ is λ for every positive m: it pushes with the same force all the way to 0.

8. Ridge or Lasso?

> **Key point:** Many features, some probably useless: Lasso. Most features useful, especially correlated ones: Ridge.

	Ridge	Lasso
Penalty	squares, $\lambda \sum \beta_j^2$	absolute values, $\lambda \sum \lvert\beta_j\rvert$
Coefficients	shrink, never exactly 0	shrink, many become exactly 0
Feature selection	no	yes
Best when	most features matter	only some features matter
Closed-form formula	yes	no (solved step by step)

The "best when" row follows ISL §6.2.2: Lasso does a little better when many features are useless, Ridge when most are useful (also StatQuest, "Regularization Part 2"). Lasso has no closed-form formula because the absolute values make the answer non-linear in y (ESL §3.4.2). With strongly correlated features, Lasso tends to keep only one of the group, and Ridge has been seen to predict better (Zou and Hastie 2005, §1).

Figure 9 puts the two side by side on the diabetes split, both with alpha 0.1. Ridge keeps all ten coefficients non-zero; Lasso sets age, s2 and s4 to exactly 0 and pays only 0.02 of test R^2 for it.

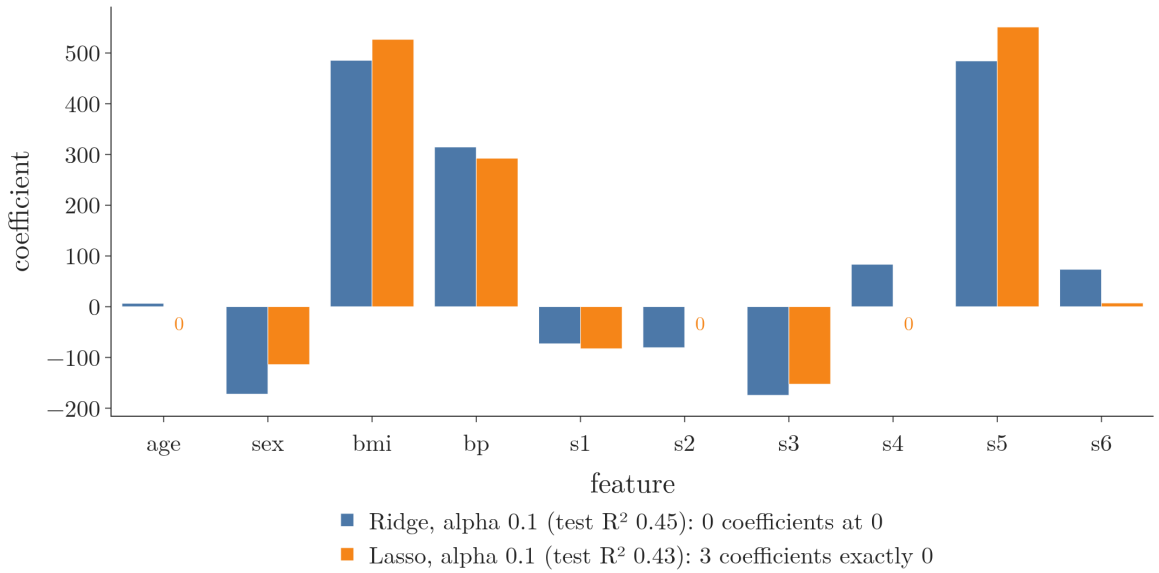


Figure 9: Coefficients on the diabetes split with alpha 0.1. Ridge (blue, test R^2 0.45) keeps all ten features; Lasso (orange, test R^2 0.43) sets age, s2 and s4 exactly to 0.

Both shrink the largest coefficients, raise bias, lower variance and are tuned through λ. The difference that matters in practice is that Lasso can remove features. **Elastic Net** (G-667; a later Note) combines both penalties.

> **Python:** Lasso in scikit-learn.
> ```
> from sklearn.linear_model import Lasso
> ```

```
lasso = Lasso(alpha=0.1)
lasso.fit(X_train, y_train)
lasso.coef_                      # several exactly 0
lasso.score(X_test, y_test)      # R^2 0.43 on the diabetes split
```

As with Ridge, features should be on the same scale first, for example with `make_pipeline(StandardScaler(), Lasso(alpha=0.1))`, because the penalty depends on each coefficient's size, and that size depends on its feature's units (why standardise before Ridge).

9. Summary

- Lasso adds $\lambda \sum |\beta_j|$ to the squared error: **L1 regularisation** (G-1026), so large coefficients are penalised by their size, as in Ridge, but without squaring.
- As λ grows, coefficients shrink and then become exactly 0, one by one; size alone does not protect a coefficient, because the large s2 goes second.
- Zero coefficients remove features: Lasso performs feature selection (diabetes: alpha 1 keeps bmi, bp and s5; the polynomial: alpha 0.1 keeps x and x^2), so it suits data with many columns where only some matter.
- A larger λ means more bias and less variance; too large removes every feature, so pick λ in between (alpha 0.1 had the lowest test error on the polynomial).
- The absolute value makes a corner at 0 in the loss, which is why the answer can sit exactly at 0; Ridge's smooth penalty has no corner, so its slope only approaches 0.
- So Lasso is the regularised linear regression to choose when you also want it to drop useless features while it trains.

10. Sources

Built from

- CampusX, "Lasso Regression | Intuition and Code Sample | Regularized Linear Models", YouTube, https://www.youtube.com/watch?v=HLF4bFbBgwk
- StatQuest with Josh Starmer, "Regularization Part 2: Lasso (L1) Regression", YouTube, https://www.youtube.com/watch?v=NGf0voTMlcs. The useless-variables example (Section 4) and the Ridge-or-Lasso guidance (Section 8).
- StatQuest with Josh Starmer, "Ridge vs Lasso Regression, Visualized!!!", YouTube, https://www.youtube.com/watch?v=Xm2C_gTAl8c. The side-by-side loss curves (Figure 8), redrawn on our own data.

Other references

- **Tibshirani 1996:** Tibshirani, R. "Regression Shrinkage and Selection via the Lasso." *Journal of the Royal Statistical Society B* 58(1), 267–288, 1996. Section 1.
- **ISL:** James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*, 2nd ed. Springer, 2021. Section 2.2.2, p. 34; Section 6.2.2, pp. 242, 246.
- **ESL:** Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Sections 3.4.1 (p. 63) and 3.4.2 (p. 68).
- **Zou and Hastie 2005:** Zou, H. and Hastie, T. "Regularization and Variable Selection via the Elastic Net." *Journal of the Royal Statistical Society B* 67(2), 301–320, 2005. Section 1.
- **scikit-learn docs:** `sklearn.linear_model.Lasso`, scikit-learn 1.9.

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Lasso regression (G-1047)	Linear regression with a penalty on the sum of absolute coefficients (L1); it shrinks coefficients and can set some exactly to 0, which removes those features.
L1 regularisation (G-1026)	Another name for the absolute-value penalty used by Lasso.
Sparse model (G-1844)	A model in which many coefficients are exactly 0; those features have no effect and can be dropped, so the model also does feature selection.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Target (G-1949)	The output column we predict, such as the class; also called the label.
Feature selection (G-768)	Keeping only the useful input columns and dropping the rest.
R^2 score (coefficient of determination) (G-1717)	1 minus the model's squared error divided by the squared error of always predicting the mean: 1 is perfect, 0 is no better than the average.
Overfitting (G-1429)	Learning the training data too closely, noise included; fails on new data.
Underfitting (G-2035)	Being too simple to capture the pattern; fails on all data.
Elastic Net regression (Elastic Net) (G-667)	Linear regression whose loss gets both the L2 (ridge) and the L1 (lasso) penalty, each with its own strength. It shrinks the coefficients and can set some to 0, so we need not choose between ridge and lasso in advance.